



Andrographis paniculata improves growth and non-specific immunity of shrimp *Litopenaeus vannamei*, and protects it from *Vibrio alginolyticus* by reducing oxidative stress and apoptosis

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ABSTRACT

Andrographis paniculata (AP) is a traditional medicinal plant with many pharmacological activities, including anti-inflammatory, antimicrobial, immunity stimulation and so on. Several studies have reported that AP plays a strong role in promoting the immune system of aquatic animals to resist several pathogens. In the present study, we investigate the effects of a diet containing AP on the immune responses, growth, and the resistance to *Vibrio alginolyticus* (*V. alginolyticus*) in *Litopenaeus vannamei* (*L. vannamei*). Four diets were formulated by adding AP at the dosage of 0% (Control), 0.25%, 0.5%, and 1% in the basal diet, respectively. Each diet was randomly fed to one group with three replicates of shrimps in a 28-day feeding trial. The results showed that dietary AP improved the growth performance and non-specific immune function of shrimps. To investigate the effect of AP on disease resistance of *L. vannamei*, shrimps fed with diet containing AP were challenged with *V. alginolyticus*. Compared with the control group, the shrimps fed diet containing AP showed significantly higher survival. Furthermore, the hepatopancreas injury in the shrimp fed with AP was less than control group at 6 h after *V. alginolyticus* infection. However, no difference was observed in the degree of hepatopancreas injury between AP groups and control group at 12 h and 24 h after *V. alginolyticus* infection. Based on this result, the samples at 6 h after *V. alginolyticus* infection was selected for subsequent detection. Reactive oxygen species (ROS) accumulation in hemocytes and O_2^- production in hepatopancreas caused by *V. alginolyticus* infection was significantly reduced after feeding a diet containing 0.25% and 0.5% AP ($p < 0.05$). In addition, we found that feeding AP significantly up-regulated the expression of pro-apoptotic genes (Bax, Caspase 3, p53) and down-regulated the expression of anti-apoptotic genes (Bcl-2) in hepatopancreas after *V. alginolyticus* infection. In conclusion, AP promote the growth and immunity of *L. vannamei*, and protects shrimps against *V. alginolyticus* by regulating the oxidative damage and apoptosis. These results provide useful information regarding the effects of AP extracts as a shrimp feed additive for sustainable shrimp culture.

1. Introduction

The white shrimp, *Litopenaeus vannamei* (*L. vannamei*), is one of the most important commercial mariculture shrimp species in the world and in China (Li and Xiang, 2013). With the continuous expansion of shrimp culture, the shrimp culture industry faces serious adverse impacts of deteriorating pond environments, intensified cultivation, and viral and bacterial epidemics (Lu et al., 2017; Soto Rodriguez et al., 2010).

Though the use of chemicals and antibiotics reduce the occurrence of disease, but threaten the environment and human health. Therefore, it is important for the shrimp aquaculture industry to develop methods to effectively prevent the spread of viral and bacterial diseases. In 2019 China banned the use of feed additives for all growth-promoting drugs except Chinese herbal medicine in food animals (Yu et al., 2021). Chinese herbal medicines have been known as immunostimulants for thousands of years (Tan and Vanitha, 2004). The immunostimulatory

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activity of herbal components has been most widely studied in mice (Amani et al., 2021), chickens (Ullah et al., 2018) or human cell lines (Choi et al., 2014).

In aquaculture, many medicinal plants are also used as immunostimulants and feed additives. They have the properties of growth promoting ability, improve the immune system, antimicrobial capability, and stimulating appetite and anti-stress characteristics (Prabu et al., 2018; Vaseeharan et al., 2011; Xie et al., 2018). *Andrographis paniculata* (Burm.f.) Nees (AP) is an herbaceous plant widely distributed in Southern Asia, which is commonly known as the “King of Bitters” in Acanthaceae family. In modern pharmacology, AP has been used to stimulate the immune system and treat myocardial ischemia, pharyngotonsillitis, and respiratory tract infection (Akbar, 2011; Hossain et al., 2014). AP also exhibits anti-microbial (Guo et al., 2014), anti-inflammatory (Li et al., 2017), anti-hyperglycemic (Thakur et al., 2016), oxygen radical scavenging (Sivakumar and Rajeshkumar, 2015), anti-atherosclerotic (Al Batran et al., 2014), and anti-cancer (Banerjee et al., 2016; Mi et al., 2016) activities (Sukardiman et al., 2007) in vivo and in vitro. Several studies have reported that AP plays a strong role in promoting the immune system of aquatic animals to resist several pathogens. A study found that diets supplemented with AP leaf powder or dried matter of AP aqueous extract increased the survival rate of *Nile tilapia* after challenge with *S. agalactiae* (Rattanachaikunsopon and Phumkhachorn, 2009). A mixture of AP and other Chinese herbs also increased growth of shrimp and boost the immune response of *P. monodon* against *V. harveyi* (Aftabuddin et al., 2017). Although AP has been widely used in human diseases, there is little research on its application in aquatic animals, especially in shrimp disease prevention. Therefore, the safety of AP in *L. vannamei* was evaluated through an investigation of the effect of this dietary supplement on the nonspecific immune response, growth performance, and disease resistance against *V. alginolyticus* in *L. vannamei*. The result of the present study provides useful information regarding the effects of AP extracts as a shrimp feed additive for sustainable shrimp culture.

2. Materials and methods

2.1. Preparation of AP

AP, collected from Xi'an city (Shanxi Province, China), were purchased from Xi 'an Huilin Biological Technology Co., LTD (Xi 'an, China) in August 2020. 1 kg dried AP were milled into a fine powder and mixed with hot distilled water (98 °C) at a ratio of 1:9 and subsequently left for 24 h at room temperature. The suspension was boiled at 100 ± 2 °C for 1 h. The resulting solution was filtered using a muslin cloth and Whatman filter paper no. 1 to separate the solids and the liquid. For the lyophilization, the liquid was frozen at -80 °C and dried in a freeze dryer for three days to yield 100 g extract powder. The powder extract was stored at -20 °C until further use.

2.2. Diet preparation

A high-quality, practical diet was used as control and 3 experimental diets based on control diet were supplemented with different concentrations of AP, to obtain a final concentration of 2.5 g, 5 g, and 10 g AP per kg of feed (named as 0.25% AP, 0.5% AP and 1% AP). Basal diet was purchased from Guangdong Shuyuan Aquatic Science and Technology Co., LTD. Pulverize the basal diet with a grinder and sieve it using a nylon sieve (300 mesh size). The experimental diets were prepared by mixing the basal diet pulverized powder and different concentrate of AP, then adding 40% water until a stiff dough was obtained. The dough was pelleted by double-screw extruders (F-26 [III], China) with diameters of 1.0 mm. The difference of AP supplemental amount in each group was replaced by the same amount of basal feed pulverized powder.

2.3. *Litopenaeus vannamei*

Healthy shrimps (*L. vannamei*) were collected from an aquaculture farm in Panyu, Guangdong Province, China. Before the experiment, the shrimps for this experiment were tested for a number of common pathogens (WSSV, IHHNV, EHP, AHPND, CMNV, DIV1, IMNV, YHV, TSV) to ensure health prior to the growth and challenge test. Shrimps were acclimatized in filtered aerated seawater at constant temperature 24 ± 1 °C and salinity of 10 ± 1 ppt, 7 days prior to the experiment, and fed daily with a ratio of 5% of body weight using commercial feed (Shuyuan Aquatic Science and Technology Co., LTD, Guangdong, China). Shrimps in the intermolt stage (stage C) were selected for the experiments. A total of 360 tails *L. vannamei* shrimps (weight of 2.311 ± 0.105 g) were randomly distributed into twelve 100 L glass aquaria containing 80 L of sea water (randomly assigned 30 tails/aquarium). The experiment was divided into four groups (control, 0.25% AP, 0.5% AP, 1% AP) with three duplicates per group. The shrimp were fed three times per day, at 08:00, 14:00, and 20:00. The ration was initially 5–7% of body weight and was adjusted thereafter according to the feeding response of the shrimp in each tank. The feeding experiment lasted for 28 days.

2.4. Total hemocyte count (THC)

Shrimp hemolymph samples from the AP groups and control groups were collected from the ventral side of the second abdominal segment using a 1 mL sterile syringe with 500 µL of a modified anticoagulant. Mixed hemolymph samples were placed on a hemocytometer and the THC were measured under an optical microscope (Olympus CX31RBSF, Beijing, China).

2.5. Shrimp growth

After the 28-day feeding trial, all shrimp were fasted for 24 h and then weighed to calculate weight gain (WG, %), specific growth rate (SGR, %), Feed conversion ratio (FCR) and survival rate (SR, %) according to the following equations:

Weight gain (WG, %) = [(Final weight - Initial weight) / Initial weight] $\times$ 100;

Specific growth rate (SGR, %) = (Ln Final weight - Ln Initial weight / Experimental days) $\times$ 100;

Feed conversion ratio (FCR) = Feed intake (FI) / (Final weight (g) - Initial weight (g));

Survival rate (SR, %) = Number of final shrimps / Number of initial shrimps $\times$ 100;

2.6. Challenge test

The *V. alginolyticus* was cultured overnight at 37 °C using Luria-Bertani (LB) medium. Then, bacteria solution was harvested and washed three times with sterilized phosphate-buffered saline (PBS) by centrifugation at 3000 rpm for 10 min. After the 28-day feeding regime, 40 shrimps from each treatment were challenged with *V. alginolyticus* by injecting 10 µL of bacterial suspension into the ventral sinus of the shrimp using microinjector at a dose of 10^7 colony forming units (CFU) ml⁻¹. While, the shrimps that received no AP and received *V. alginolyticus* suspension (10 µL) or only PBS (10 µL) performed as the challenged control and unchallenged control, respectively. Shrimps were fasted 24 h before trial to avoid interference of diet. Experimental shrimps were kept into ten plastic aquariums ($50 \times 38 \times 30$ cm, randomly assigned 20 tails/aquarium). Each aquarium was provided with continuous aeration during the challenge test. Three shrimps in

Table 1
Primers used for PCR amplification.

Name	Sequence (5'–3')
For RT-qPCR	
RT-LvLSZ-F	TTCCGATCTGATGTCGGATGA
RT-LvLSZ-R	TTGCTGTTGTAAGCCACCCAG
RT-LvPO-F	GAAGTCCATTCGTCGCTCTG
RT-LvPO-R	CGGCTTCGCTCTGGTTAGG
RT-LvSOD-F	GCGTTGGAGTGAAAGGCTCT
RT-LvSOD-R	TCACGTAATCTGCACGGAGG
RT-LvCrustin-F	ATTCTGTGCGGCTCTTTAC
RT-LvCrustin-R	ATCGGTCTGTTCTTCAGATGG
RT-LvPEN3-F	GGGTTTCCTCCTGCGTCCG
RT-LvPEN3-R	GAAGGGTGGTGGCTGGGT
RT-LvALF1-F	ACAGGCTTCCGAGCAACAC
RT-LvALF1-R	TGGACAAGAGCAATCAGG
RT-LvCAT-F	GAGGCCGTCTACTGCAAGTT
RT-LvCAT-R	GAGGGGTAATCGCCACTGA
RT-LvGPX-F	AGGGACTTCCACCATG
RT-LvGPX-R	CAACAACTCCCTTCGGTA
RT-LvGST-F	AAGATAACGCAGAGCAAGG
RT-LvGST-R	TCGTAGGTGACGGTAAAGA
RT-LvP53-F	CGAATCCCAACATCCACG
RT-LvP53-R	GGCGGCTGATACACCACC
RT-LvBax-F	GGTGAATCACAAGAGAGCGA
RT-LvBax-R	TGTTCTCCACGGTGTCTCAC
RT-LvBcl-2-F	CCTTGCTTGACACAGTCCGA
RT-LvBcl-2-R	CAGACAAGGTCGTGAGGTGG
RT-LvCaspase3-F	AGTTAGTACAAACAGATTGGAGCG
RT-LvCaspase3-R	TTGTGGACAGACAGTATGAGGC
RT-LvAmy-F	TCAGGTATCACCGCCTAACG
RT-LvAmy-R	CGGAGCGAGTGACGAGTTT
RT-LvTry-F	AGGGCGAGGACATGAACAAC
RT-LvTry-R	GCTGATGGTGAAGCCGTTG
RT-Lvβ-actin-F	GCCCATCTACGAGGGATA
RT-Lvβ-actin-R	GGTGGTCGTGAAGGTGTAA

each aquarium were sampled at 0 h, 6 h, 12 h and 24 h after *V. alginolyticus* injection. Hemolymph and hepatopancreas was extracted and reserved. The SR was calculated using the following formula:

$$\text{SR (\%)} = (\text{Number of shrimps that survived}) / (\text{Number of shrimps that injected}) \times 100$$

2.7. Total RNA isolation and quantitative real-time PCR (qRT-PCR)

Total RNA was extracted from hepatopancreas tissues of shrimps using Trizol (Vazyme Biotech, Nanjing, China) following the manufacturer's instructions. The total RNA (1 µg in each reaction) was used as a template to synthesize first-strand cDNA with HiScript III 1st Strand cDNA Synthesis Kit (+gDNA wiper) (R312-01, Vazyme, Nanjing, China) according to the manufacturer's instructions. qRT-PCR was carried out in an ABI 7500 T thermal Cycler (Applied Biosystems, Foster City, CA) using TransStart® Top Green qPCR SuperMix (AQ131-01, TransGen Biotech, Beijing, China) following the manufacturer's recommendations. qRT-PCR amplification was carried out as follows: 95 °C for 5 min, followed by 40 cycles at 95 °C for 10 s and 60 °C for 30 s. All primers (Table 1) were further verified by sequencing after the PCR products. The relative values of gene expression levels were expressed as $2^{-\Delta\Delta C_t}$. All experiments were repeated three times.

2.8. Phagocytic activity assay

To investigate the phagocytic activity of hemocyte in *L. vannamei*, hemocytes were adjusted to 1×10^7 cells/mL using Leibovitz's L-15 medium supplemented with 10% FBS. Hemocytes were incubated with 0.5 µm microspheres (Red beads) (Huge biotechnology, Shanghai, China) at a 1:20 (cells: beads) ratio for 4 h at 28 °C, respectively. After incubation, the cells were collected and centrifuged at $100 \times g$ for 10 min

at 4 °C to remove excess beads. Then the cells were resuspended in 1 mL PBS. After three times washes with PBS, the phagocytic activities of hemocytes from each group were independently analyzed by using BD Arial III flowcytometer (BD, USA). Hemocytes incubated without microspheres were also included as blank controls. Data analyses were performed using FlowJo X. Phagocytic activities of hemocytes were expressed as phagocytic rate and phagocytic index, were calculated as:

$$\text{Phagocytic rate (\%)} = (\text{Phagocytic hemocytes}) / (\text{Total hemocytes}) \times 100$$

$$\text{Phagocytic index} = (\text{Beads per phagocytic cell}) / (\text{Number of phagocytic cells})$$

2.9. Antioxidant enzyme detection

The hepatopancreas tissues of shrimps were accurately weighed, and 9 times the volume of PBS was added according to 1:9. Mechanical homogenization was performed under ice water bath, and centrifugation was conducted for 10 min at 2500 rpm. Total protein was quantified using BCA Protein Quantification Kit (Yeasten, Shanghai, China). Superoxide dismutase (SOD), catalase (CAT), total antioxidant capacity (T-AOC) and malondialdehyde (MDA) were detected using relevant assay kits (Beyotime Biotechnology, Shanghai, China) following the manufacturer's instructions.

2.10. Histopathology

The hepatopancreas tissue of *L. vannamei* were immobilized in 10% formalin for 24 h and then dehydrated in increments of alcohol (50%–95%). The dehydrated tissues were embedded in paraffin and cut into 4 µm thick slices. Tissues sections were stained with hematoxylin and eosin (H&E), then examined using a light microscope (DM6, Leica, Germany).

2.11. Determination of reactive oxygen species (ROS) generation in hemocytes and O_2^- production in hepatopancreas

To measure the production of ROS by hemocytes, the cell-permeant probe 2',7'-dichlorofluorescein diacetate (DCFH-DA) kit (Beyotime, Shanghai, China) was used according to the manufacturer's instructions. The hemolymph and anticoagulant mixture were incubated in the dark for 30 min with 10 µM DCFH-DA. Then, the fluorescence of the cell suspensions was recorded using flow cytometry. ROS production is represented by the mean DCF fluorescence. For each sample, 10,000 events were collected.

About 0.1 g of shrimp hepatopancreas tissue was weighed, 1 mL of extract was added, and homogenized in an ice bath. Then 10000 g was centrifuged at 4 °C for 20 min, and the supernatant was placed on ice for testing. The O_2^- production of hepatopancreas was quantified using oxygen free radical (OFR) Kit (Mlbio, Shanghai, China).

2.12. TUNEL assay

Apoptosis was detected by TUNEL assay kit (Elabscience, Wuhan, China). Hepatopancreas tissue sections were deparaffinized in 1X Dewaxing solution at 60 °C for 50 min and then incubated with 1X proteinase K for 20 min at 37 °C. After washing three times with PBS buffer, the sections were incubated in TdT equilibration buffer for 30 min, then incubated with 50 µL TUNEL reaction mixture containing 35 µL of TdT equilibration buffer, 10 µL of AF488-labeling solution and 5 µL of TdT Enzyme for 60 min at 37 °C in the dark. Then, the slides were rinsed in PBS three times and nuclei were counterstained with 4, 6-diamidino-2-phenylindole dihydrochloride (DAPI) for 5 min at room temperature in the dark. The specimen was covered in streptavidin-HRP for 30 min and washed three times with PBS. The slides were rinsed four times with PBS and observed under microscope (DM6, Leica, Germany).

Table 2

The initial weight, final weight, weight gain (WG, %), specific growth rate (SGR, %), food conversion rate (FCR), and survival rate (SR, %) of *L. vannamei* fed different concentrations AP diet for 28 days.

Diet	Initial Weight (g)	Final Weight (g)	WG (%)	SGR (%)	FCR	SR (%)
Control	2.239 ± 0.096	6.028 ± 0.777	159.979 ± 13.392	3.181 ± 0.173	1.411 ± 0.122	83.333 ± 4.409
0.25% AP	2.288 ± 0.140	7.303 ± 0.658*	218.844 ± 12.461*	3.863 ± 0.129*	1.065 ± 0.106**	83.333 ± 1.667
0.5% AP	2.306 ± 0.154	7.193 ± 0.456*	212.045 ± 4.924*	3.793 ± 0.053*	1.083 ± 0.080**	93.333 ± 3.333*
1% AP	2.277 ± 0.069	7.274 ± 1.535**	220.013 ± 71.874**	4.189 ± 0.749**	0.930 ± 0.379**	77.500 ± 12.500

Note: Data are presented as mean ± SD. * $p < 0.05$, ** $p < 0.01$.

2.13. Statistical analysis

Significantly differences between different groups were analyzed by one-way analysis of variance (ANOVA) using SPSS software version 18.0 (IBM, Armonk, NY, USA). Duncan's multiple-range test was used for the multiple comparisons of group means. All graph were implemented in GraphPad Prism V8.0.2 (GraphPad, San Diego, CA, USA). All of the experiments described in this study were conducted in triplicates, and data was presented as the mean ± standard error. Difference was considered as significant at p value < 0.05 .

3. Results

3.1. Effect of dietary AP on growth performance

Growth performance indices of *L. vannamei* fed with AP-supplemented diets for 28 days are presented in Table 2. Shrimps fed with different levels of dietary AP showed significant ($p < 0.05$) increase in final weight, WG (%), SGR (%), and FCR particularly at 0.25% and 0.5% AP treatments. The second-order polynomial regression between final weight (g), SGR (%), WG (%), and FCR and dietary AP levels (Fig. 1) shows that the optimal levels of dietary AP for *L. vannamei* are 0.25–0.5%. Among the treatment groups, 0.5% AP group exhibited the highest SR (%) ($93.333 \pm 3.333\%$), were significantly higher than the other group ($p < 0.05$) (Table 2).

3.2. Effect of AP on the digestive performance of shrimp

We detected the expression of amylase (Amy) and trypsin (Try) in hepatopancreas and intestinal tract by qPCR. The results showed that feeding AP could improve the digestive performance of *L. vannamei* (Fig. 2). Fed diet with 0.25% AP significantly increased the expression of Amy and Try whether in hepatopancreas or in intestinal tract (Fig. 2 A–D, $p < 0.05$). However, there was no significant effect on the digestive performance of *L. vannamei* when fed diet with 1% AP (Fig. 2 B–D, $p > 0.05$).

3.3. Effects of AP on *L. vannamei* immunity parameters over a 28-day feeding regime

3.3.1. THC

The THC of shrimps was detected after fed a diet containing AP 14 and 28 days, the results showed in Fig. 3. Compare with the control group, the THC was significantly increased in the group of 0.25% AP and

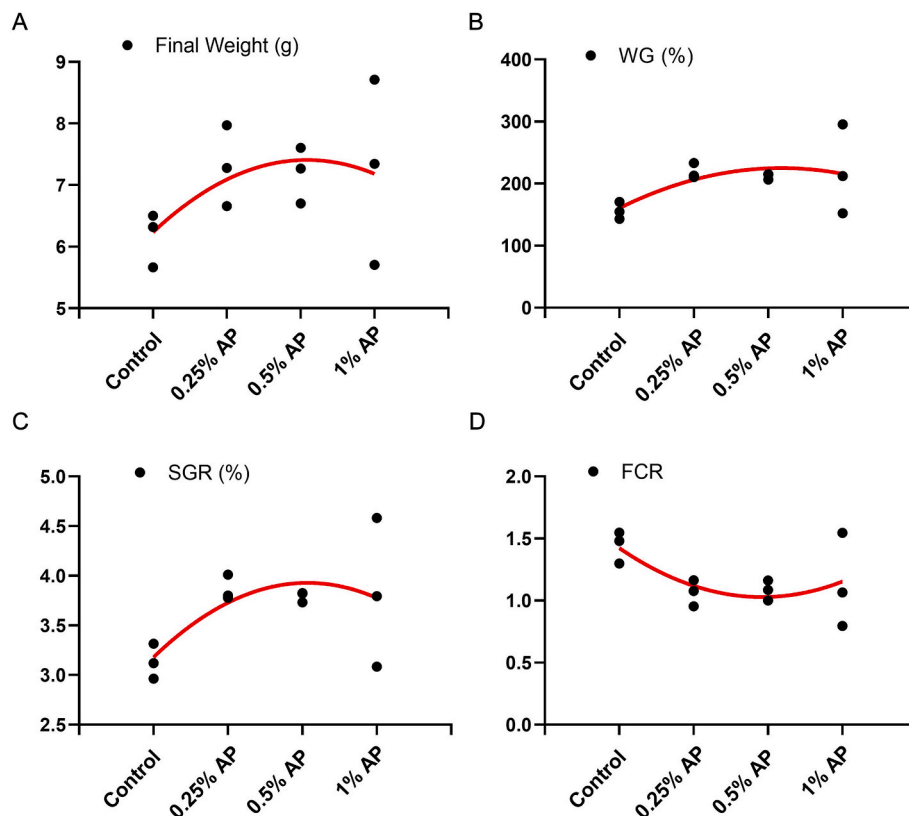


Fig. 1. The relationship between final weight (g) (A), specific growth rate (SGR; %/day) (B), weight gain (%) (C), and feed conversion ratio (FCR) (D) of *L. vannamei* and various levels of AP extract.

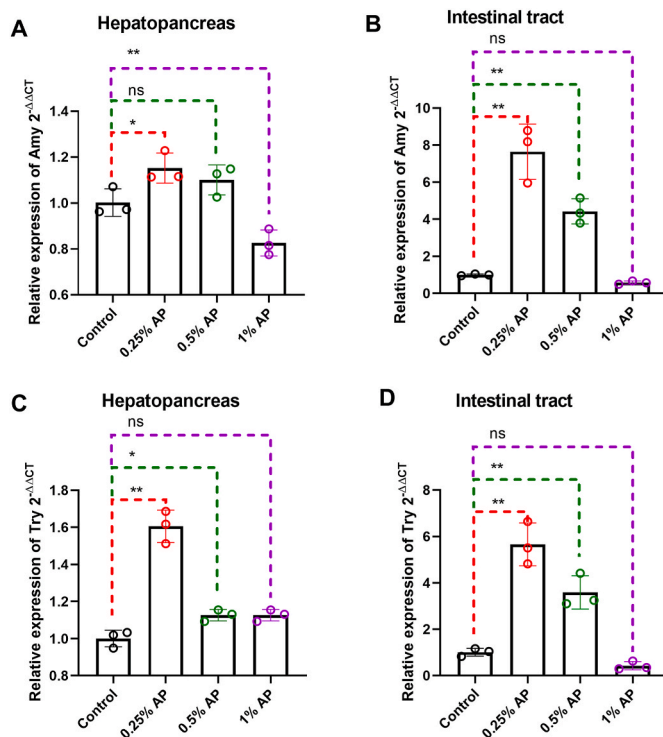


Fig. 2. Effects of AP on expression of amylase (Amy) and trypsin (Try) in hepatopancreas and intestinal tract of *L. vannamei* (A–D). Results were shown as mean $\pm$ SD. Statistical differences were evaluated by a one-way ANOVA, where ns means $p > 0.05$, *means $p < 0.05$, **means $p < 0.01$.

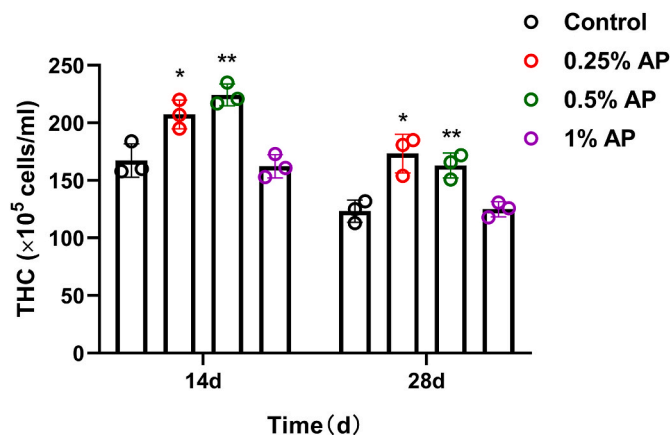


Fig. 3. Total hemocytes count (THC) of *L. vannamei* fed a diet containing 0.25%, 0.5%, or 1% of AP for 14 and 28 days. Data are presented as mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$.

0.5% AP ($p < 0.05$). However, no significantly difference was observed between 1% AP group and control group whether feeding 14 days or 28 days.

3.3.2. Phagocytic activity

The results of phagocytosis detected by flow cytometry were shown in Fig. 4A. After feeding shrimps containing 0.25% AP and 0.5% AP, the phagocytic rate of hemocytes were $35.033 \pm 0.751\%$ and $35.233 \pm 1.106\%$, and the phagocytic index of hemocytes were 1.026 ± 0.031 and 0.886 ± 0.025 , respectively. Compared with control group (phagocytic rate: $21.233 \pm 0.777\%$, phagocytic index: 0.707 ± 0.004 , $p < 0.05$), phagocytic rate and phagocytic index in the group of 0.25% AP and 0.5% AP were significantly elevated ($p < 0.05$). However, no

significantly difference was observed between 1% AP group with control group (Fig. 4B and C).

3.3.3. Expression of innate immune-related genes

To further explore the role of AP on immunity of *L. vannamei*, expression of a series of immune related genes after fed the AP for 28 days in vivo was investigated using RT-qPCR method, shown in Fig. 5. Feeding diets supplemented with AP, expression of almost all the examined immune effector genes involved in humoral immunity, including immune functional proteins (LSZ, PO and SOD), antimicrobial peptides (ALF1, PEN3 and crustin) and antioxidant protein (GST, GPX and CAT) significantly increased (Fig. 5A–I). When the diets containing 0.25% AP, the expressions of LSZ, PO, SOD, ALF1, PEN3, crustin, GST, GPX and CAT in hepatopancreas of *L. vannamei* reached the highest, were 3.663 ± 0.074 , 1.406 ± 0.143 , 3.914 ± 0.179 , 1.862 ± 0.060 , 8.313 ± 0.448 , 2.994 ± 0.471 , 2.196 ± 0.581 , 2.558 ± 0.086 and 4.097 ± 0.132 (compare with control group $p < 0.01$).

3.3.4. Antioxidant enzymes

The antioxidant enzymes forming the first line of defense, SOD, T-AOC and MDA play a vital role countering ROS-driven oxidative stress (Fig. 6). The SOD activity in hepatopancreas of shrimps fed AP diets containing 0.25% and 1% was significantly higher than that in control group ($p < 0.05$). However, no significantly was observed between 0.5% AP group and control group ($p > 0.05$). The T-AOC in hepatopancreas of *L. vannamei* fed diets containing AP were significantly higher than those in the control group ($p < 0.01$), and the contents of T-AOC in hepatopancreas reached the highest (39.181 ± 0.204 mM/mg protein) when the supplemental level of AP was 0.5%. Whereas the highest MDA level in the control group compared to the other treatments ($p < 0.05$). The lowest MDA level was observed in the 0.25% AP group (8.390 ± 0.102 mM/mg protein) compared to the other group ($p < 0.05$).

3.4. Effect of AP on bacterial disease resistance

The cumulative mortality increased over time in all groups after challenging with *V. alginolyticus* (Fig. 7). The *V. alginolyticus* strain we used was *V. alginolyticus* ZJ51, which was donated by South China Sea Institute of Oceanology. The strain ZJ51 was cultured in TCBS plates and single colony was picked out for subsequent experiments. Compared to the control + *V. alginolyticus* group, the cumulative mortality of shrimps in the AP + *V. alginolyticus* group decreased after *V. alginolyticus* infection 6 h. The survival rates for the control + *V. alginolyticus* group and 0.25%, 0.5%, 1% AP + *V. alginolyticus* group were 33.3%, 77.7%, 72.2% and 44.4% at 6 h after *V. alginolyticus* infection, respectively. At 12 h after *V. alginolyticus* infection, the survival rate of shrimps in control + *V. alginolyticus* group was 0%, the death time of shrimp was significantly advanced than that in AP + *V. alginolyticus* groups.

3.5. The effect of AP on *L. vannamei* hepatopancreas under *V. alginolyticus* infection

Histopathological examination of the hepatopancreas is an important indicator of hepatopancreatic injury. Therefore, HE staining was used to detect the hepatopancreas of *L. vannamei* after *V. alginolyticus* infection 6 h, 12 h and 24 h to observe the pathological changes, which are shown in Fig. 8. After *V. alginolyticus* infection 6 h, the hepatopancreas tubules of control group were disordered, the boundary was blurred, the lumen disappeared, and some hepatic tubules were ruptured. Meanwhile, there are many inflammatory cells in interstitial spaces and tissues. Unlike the control group, the morphology of hepatopancreas tubules in shrimps fed 0.25% and 0.5% AP groups was basically normal, and the hepatic tubules were arranged orderly and the lumen was clearly visible after *V. alginolyticus* infection. However, the hepatopancreas tubule swelling and rupture occurred in the shrimp fed the 1% AP group after *V. alginolyticus* infection. The results indicated

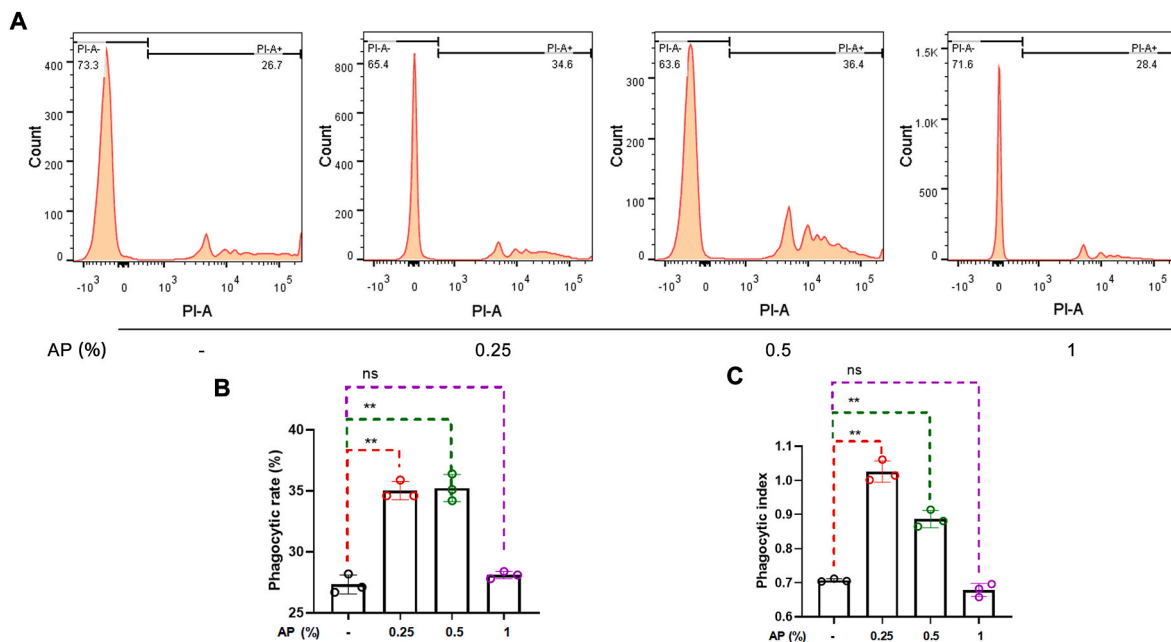


Fig. 4. Phagocytic activities of *L. vannamei* fed either a control diet or a diet containing 0.25%, 0.5%, or 1% of AP for 28 days. (A) Scatter plot and histogram of phagocytic hemocytes ingested 0.5 μ m microspheres. (B) The total average % of phagocytic cells in hemocytes. (C) The phagocytic index of phagocytic cells. Results were shown as mean $\pm$ SD. Statistical differences were evaluated by a one-way ANOVA, where ns means $p > 0.05$, *means $p < 0.05$, **means $p < 0.01$.

that adding proper concentration of AP in the feed could reduce the hepatopancreas injury caused by *V. alginolyticus* infection. At 12 h and 24 h after *V. alginolyticus* infection, there was no difference in the degree of hepatopancreas injury between AP groups and control group. Based on this result, we selected samples 6 h after *V. alginolyticus* stress for subsequent detection.

3.6. AP attenuate the oxidative damage caused by *V. alginolyticus* infection

The generation of ROS were evaluated by flow cytometry. After *V. alginolyticus* infection 6 h, ROS generation was significantly higher than that in control group (*V. alginolyticus* group vs control group: $31.27 \pm 3.67\%$ vs $22.00 \pm 2.21\%$, $p < 0.05$). Then, ROS accumulation caused by *V. alginolyticus* infection was significantly reduced after feeding a diet containing 0.25% AP ($p < 0.05$). No significantly difference was observed between 0.5%, 1% AP group and *V. alginolyticus* group ($p > 0.05$, Fig. 9 A and B).

The O_2^- production was significantly elevated to control group in shrimp after *V. alginolyticus* infection 6 h ($p > 0.05$). The O_2^- production in the hepatopancreas of the shrimp fed with 0.25%, 0.5% and 1% AP was significantly lower than that of the control group after 6 h of *V. alginolyticus* infection ($p > 0.05$, Fig. 9C).

3.7. AP reduces apoptosis of hepatopancreas induced by *V. alginolyticus* infection

The apoptosis-related gene (Bcl-2, Bax, Caspase3, p53) of hepatopancreas in *L. vannamei* after *V. alginolyticus* infection 6 h was detected by qPCR. After *V. alginolyticus* infection 6 h, pro-apoptotic genes (Bax, Caspase 3, p53) were significantly up-regulated (Fig. 10A–C, $p < 0.01$) and anti-apoptotic genes (Bcl-2) were significantly down-regulated (Fig. 10D, $p < 0.01$). The expression of pro-apoptotic genes Bax, Caspase3, p53 in hepatopancreas of shrimp fed with AP was significantly lower than that of control group 6 h after *V. alginolyticus* infection (Fig. 10A–C, $p < 0.01$). After *V. alginolyticus* infection 6 h, the expression of Bcl-2 in hepatopancreas of shrimps fed diets supplemented with 0.25% and 1% AP was significantly higher than that of control group

(Fig. 10D, $p < 0.01$), but there was no significant difference between 0.5% group and control group (Fig. 10D, $p > 0.05$). To further verify the effect of AP on the apoptosis of shrimp hepatopancreas induced by *V. alginolyticus*, TUNEL assay was conducted to detect the apoptosis of hepatopancreas in shrimp (Fig. 10E). The results of TUNEL staining were consistent with the expression of apoptotic genes in hepatopancreas.

4. Discussion

Vibriosis is regarded as the bacterial disease caused by *Vibrio* spp. in shellfish and fish aquaculture, and is also a main cause for mortality of cultured shrimp throughout the world (Lavilla-Pitogo et al., 1998). The most common treatment for bacterial diseases is antibiotics, to which most *Vibrio* spp. are usually susceptible. However, the use of antibiotics not only increases resistance to specific bacteria, but also leads to antibiotic residues in aquatic products (Ghosh et al., 2021; Li et al., 2006). In recent years, Chinese herbal feed additives are more and more favored by people (Howes et al., 2017; Wang et al., 2019, 2020; Xu et al., 2019), not only promoting health, but also preventing viral and bacterial infections by increasing the natural resistance to infection (Kaleo et al., 2019; Lim et al., 2020). Chinese herbal medicine is rich in bioactive substances, such as alkaloids, terpenoids, lectins, polyphenols, phenols, quinones and peptides, which can effectively replace antibiotics (Ina-Salwany et al., 2019). Therefore, it is of great importance to identify the role of Chinese herbal medicines in aquaculture and to develop alternative practices for growth promotion and disease management. In the present study, we evaluated the effect of AP on growth performance, non-specific immune parameters and disease resistance against *V. alginolyticus* in *L. vannamei*.

The present study revealed that AP inclusion, mainly at 0.25% and 0.5% had beneficial effects on the growth performance of *L. vannamei* (Fig. 1 and Table 2). These results may be because AP contains many bioactive compounds such as diterpenoid lactones, and flavone, which can promote the intestinal tract health and nutrient absorption of shrimp (Fig. 2). AP have demonstrated positive effects on the growth performance and health status of aquatic animals (Ngo et al., 2020). For instance, supplementing diets with methanolic extract of *A. paniculata* was reported to enhance the growth of *Oreochromis mossambicus* (Prasad

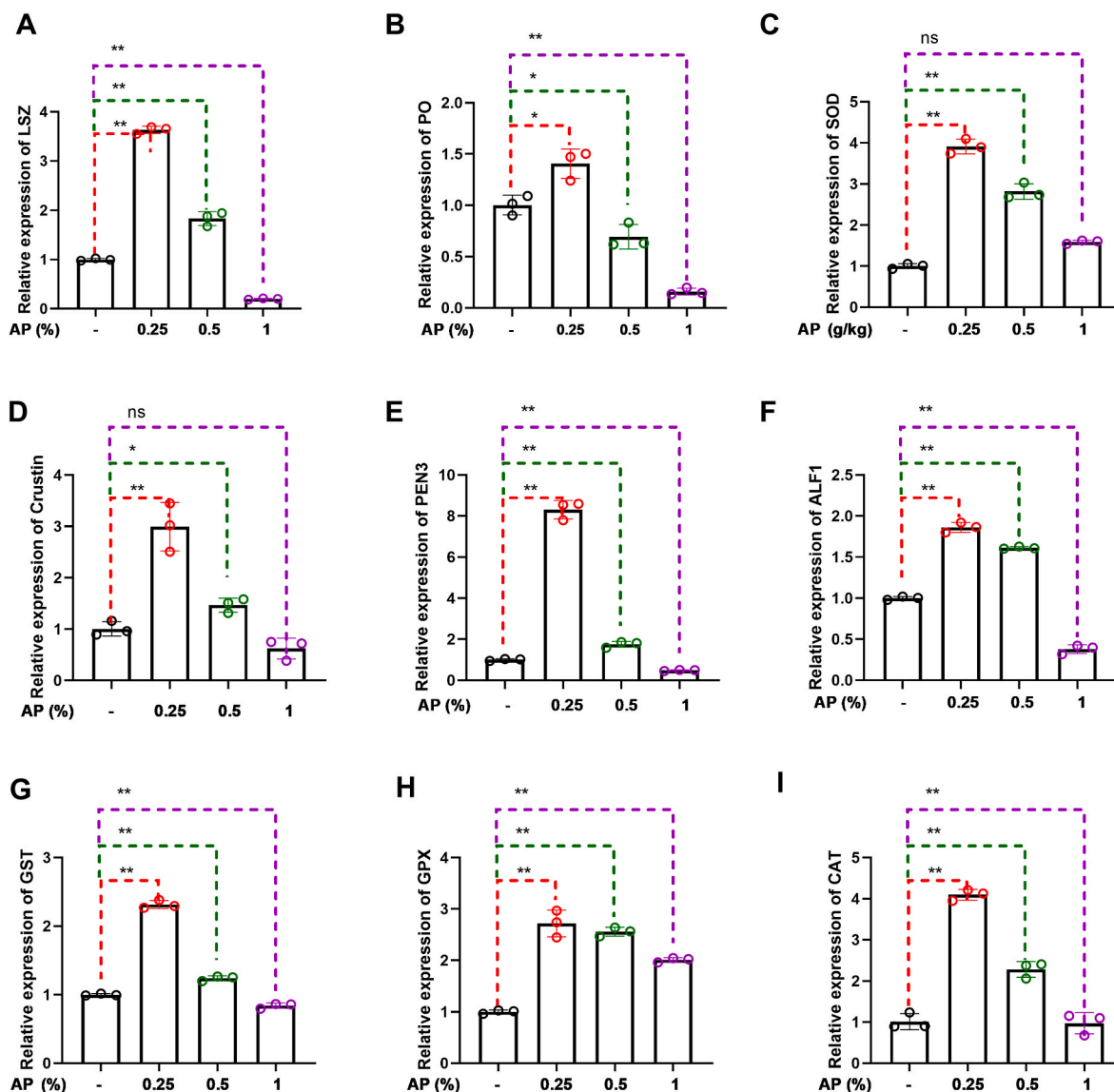


Fig. 5. Effects of AP on expression of innate immune-related genes of *L. vannamei* (A–I). Results were shown as mean $\pm$ SD. Statistical differences were evaluated by a one-way ANOVA, where ns means $p > 0.05$, * means $p < 0.05$, ** means $p < 0.01$.

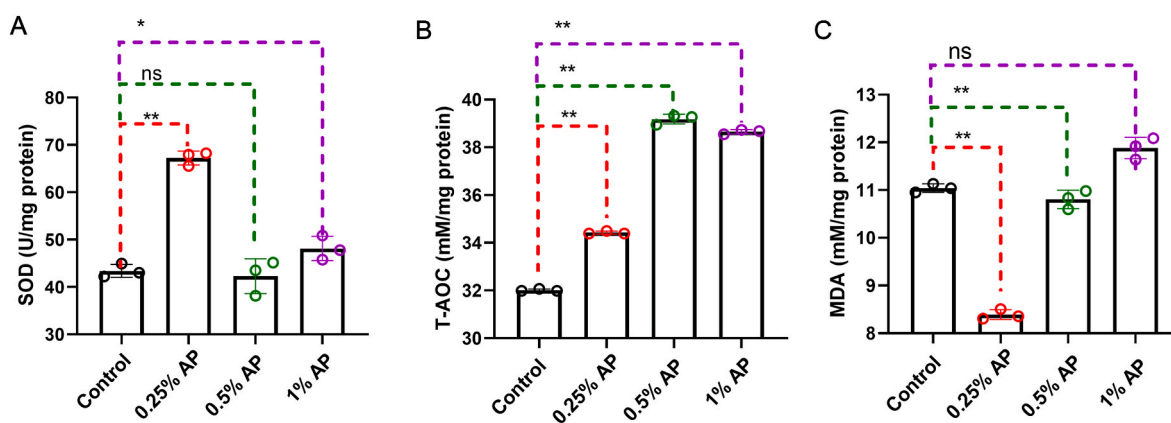


Fig. 6. SOD (A), T-AOC (B), MDA (C) activity in hepatopancreas of *L. vannamei* fed the AP for 28 days in vivo. Results were shown as mean $\pm$ SD. Statistical differences were evaluated by a one-way ANOVA, where ns means $p > 0.05$, * means $p < 0.05$, ** means $p < 0.01$.

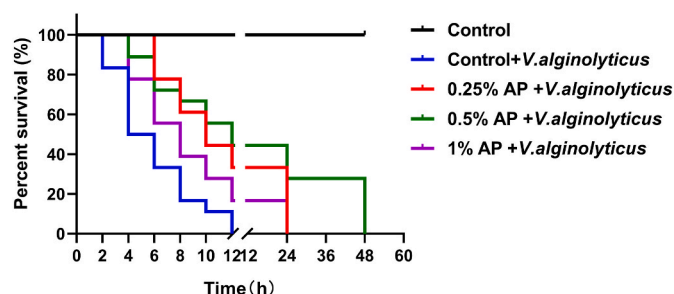


Fig. 7. Cumulative survival rate of *L. vannamei* fed either a control diet or a diet containing 0.25%, 0.5%, or 1% of AP for 28 days, and then challenged with *V. alginolyticus*.

and Mukthiraj, 2011) and *Labeo rohita* (Basha et al., 2013). The methanolic herbal extracts (*Solanum trilobatum*, AP and *Psoralea corylifolia*) helped to enhance growth and improved the health against bacterial infection in *Penaeus monodon* post larvae and AP was found to be more effective compared to other two herbal extracts (Citarasu et al., 2003; Yin et al., 2014).

Chinese herbs contain many compounds that can stimulate the immune function of fish and shrimp (Harikrishnan et al., 2011). Some active compounds could inhibit or even block the growth of a particular pathogen in the host and stimulate its immune system (Citarasu, 2010). It is worth mentioning that the use of Chinese herbal immunosuppressants before the onset of disease can reduce the loss of production (Anderson, 1992). The defense capacity of shrimps and other crustaceans is less developed than that in fish and vertebrates because they lack adaptive immune mechanisms. Thus, they apparently depend solely on innate defense systems, which include humoral and cellular components that work jointly for detection or elimination of foreign organisms and materials harmful for the host (Tassanakajon et al., 2013). Hemocytes are the primary immune cells in shrimp, and they function

through phagocytosis and encapsulation (Johansson et al., 2000). Numerous studies have demonstrated that adding Chinese herbs or their extracts can improve the immunological parameters, including THC and phagocytic activity (Kaleo et al., 2019; Liu et al., 2019; Wu et al., 2015; Yin et al., 2014). In the present study, feeding the diet containing AP can not only increase the THC (Fig. 3), but also significantly enhance the phagocytic activity of hemocytes (Fig. 4). Hemocytes are the main components of hemolymph in shrimp, which can reflect the physiological state and immune level of aquatic animals (Fan et al., 2013). Therefore, the results of THC and phagocytosis suggest that AP can enhance the immunity of *L. vannamei*, which is consistent with the expression of antioxidant and immune-related genes in hepatopancreas.

Besides the functions in digestion and metabolism, hepatopancreas also plays an important role in the innate immune response of crustaceans (Li et al., 2019). It has been shown in the shrimp *Penaeus monodon* that hepatopancreas cells can release bacterial antigens into hemolymph after phagocytic vibrio fragments (Alday-Sanz et al., 2002). In addition to the role in uptake of pathogens, the hepatopancreas is a source of molecules required for pathogen defense (Cao et al., 2014). Molecules such as phenoloxidase, lysozyme, antimicrobial peptides, and antioxidant enzymes are all participants in the innate immune response and serve to recognize, bind, and eliminate cellular pathogens or viruses (Chiu et al., 2007; Haug et al., 2002; Söderhäll and Cerenius, 1998). In the present study, phenoloxidase, lysozyme, antimicrobial peptide, and antioxidant enzyme in shrimp hepatopancreas were detected. The results indicated that adding 0.25% and 0.5% AP could increase the gene expression of phenol oxidase, lysozyme, antimicrobial peptide and antioxidant enzyme in hepatopancreas of *L. vannamei* (Figs. 5 and 6). It was similar to the effect of AP on improving non-specific immune parameters of *Labeo rohita* (Basha et al., 2013).

The improvement of non-specific immune function indicates that the shrimp's ability to resist foreign bacteria is enhanced. Many medicinal plants have been reported to increase the resistance capacity and survival of shrimp when challenged by *Vibrio* spp. (Kaleo et al., 2019; Rudi

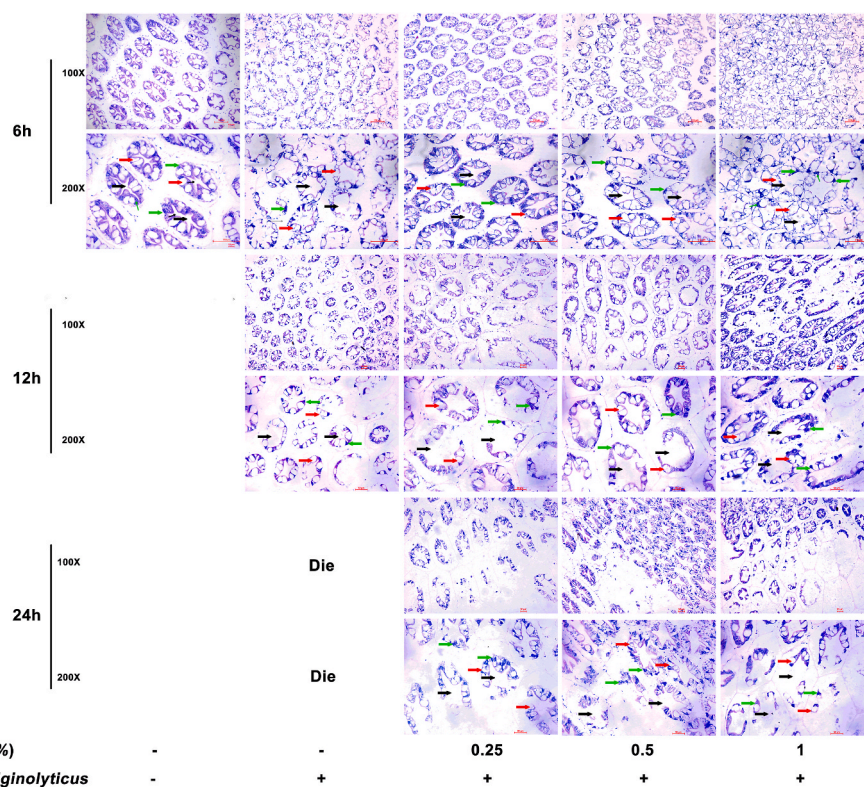


Fig. 8. AP on histopathological examination of the *L. vannamei* hepatopancreas under *V. alginolyticus* infection 6 h, 12 h and 24 h. Arrows represent lesion regions. Black arrows represent lumen of hepatic tubule of hepatopancreas, green arrows represent secretory cells of hepatopancreas, and absorptive cells of hepatopancreas.

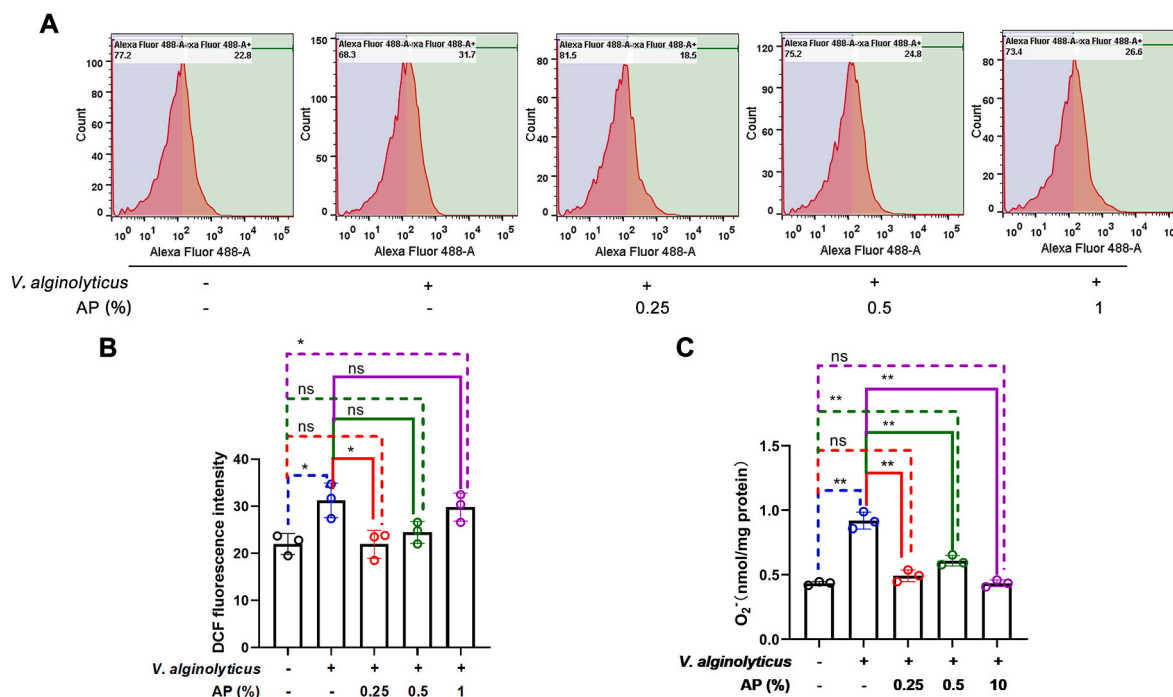


Fig. 9. AP attenuate ROS accumulation caused by *V. alginolyticus* infection. (A) Relative oxygen species level (ROS) in hemocytes of *L. vannamei* fed either a control diet or a diet containing 0.25%, 0.5%, or 1% of AP at 6 h after *V. alginolyticus* infection. (B) Column diagram of DCF fluorescence intensity. Results were shown as mean $\pm$ SD. Statistical differences were evaluated by a one-way ANOVA, where ns means $p > 0.05$, * means $p < 0.05$, ** means $p < 0.01$.

et al., 2019; Rudtanatip et al., 2017). AP plays a strong role in promoting the immune system of aquatic animals to resist several pathogens. In vivo experiments showed that fish feed supplemented with either AP leaf powder or dried matter of AP aqueous extract reduced mortality of *S. agalactiae* infected Nile tilia (Rattanachaiakunsopon and Phumkhachorn, 2009). In the present study, the treatment groups fed with dietary AP showed significantly enhanced resistance against *V. alginolyticus* infection (Fig. 7). This might be due to activation and enhancement of the non-specific immune system of experimental *L. vannamei*. Similar results in *L. rohita* fed with diets containing andrographolide showed higher relative percentage survival after challenging with *A. hydrophila* (Basha et al., 2013). In addition, adding proper concentration of AP in the feed could reduce the hepatopancreas injury caused by *V. alginolyticus* infection 6 h. However, with the prolongation of *V. alginolyticus* infection time, the hepatopancreas injury gap between AP group and control group became smaller and smaller. Based on this result, we selected samples 6 h after *V. alginolyticus* stress for subsequent detection (Fig. 8).

Previous studies have shown that *V. alginolyticus* infection can induce oxidative stress by increasing the ROS in aquaculture animals (Gu et al., 2017; Huang et al., 2018). Pathogenic bacteria-induced ROS are crucial to kill the invading bacteria in shrimp. However, the excessive production of ROS could lead to antioxidant imbalance and cell apoptosis (Davies, 2000). Thus, controlling the cellular ROS levels is conducive to the survival of cells in adverse environment (Van Erp et al., 2017). The extract of AP has been reported to have significant antioxidant activities (Venmathi Maran et al., 2022). Herein, ROS generation in hemocytes and O₂⁻ production in hepatopancreas were detected in the present study. The results showed that AP can attenuate the oxidative damage caused by *V. alginolyticus* infection (Fig. 9). Typically, the elevated ROS created by oxidative stress eventually leads to apoptosis (Gao et al., 2020). Cell apoptosis is a critical cellular event during the progression of *V. alginolyticus* infection (Wang et al., 2022; Zhao et al., 2010). The results of Gu et al. showed that *V. alginolyticus* infection influenced the apoptosis and up-regulated the expression of Caspase 3 and p53 (Gu et al., 2017; Xie et al., 2016). Inhibition of apoptosis could improve the

survival rate of *L. vannamei* under the stress of *V. alginolyticus* (Yin et al., 2022). Studies have reported that AP extract attenuates cell apoptosis (Hsieh et al., 2016; Wu et al., 2018). In the present study, feeding shrimp with AP reduced the apoptosis induced by *V. alginolyticus* infection. Moreover, the up-regulation of pro-apoptotic genes and down-regulation of anti-apoptotic genes induced by *V. alginolyticus* infection were also reversed by AP (Fig. 10).

In conclusion, the present study showed that the supplementation of AP in diets at 0.25% and 0.5% significantly enhanced the growth and promoted the non-specific immunity and the resistance to *V. alginolyticus* in *L. vannamei*. Furthermore, AP attenuated the oxidative stress and apoptosis induced by *V. alginolyticus* infection. These results provide useful information regarding the effects of AP extracts as a shrimp feed additive for sustainable shrimp culture.

CRedit authorship contribution statement

Xiaoli Yin: Conceptualization, Methodology, Software Data curation, Writing- Original draft preparation. Xueqi Zhuang & Meiqiu Liao: Visualization, Investigation, Software, Validation. Qiqian Cui & Chunxia Yan & Jiayi Huang & Zixiang Jiang: Shrimp culture, Methodology, Supervision. Lin Huang & Weitao Luo: Visualization, Investigation. Yuan Liu & Weina Wang: Writing- Reviewing and Editing.

Ethics statement

All procedures involving the handling and treatment of shrimp during this study were approved by the South China Normal University. All efforts were made to minimize suffering.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as

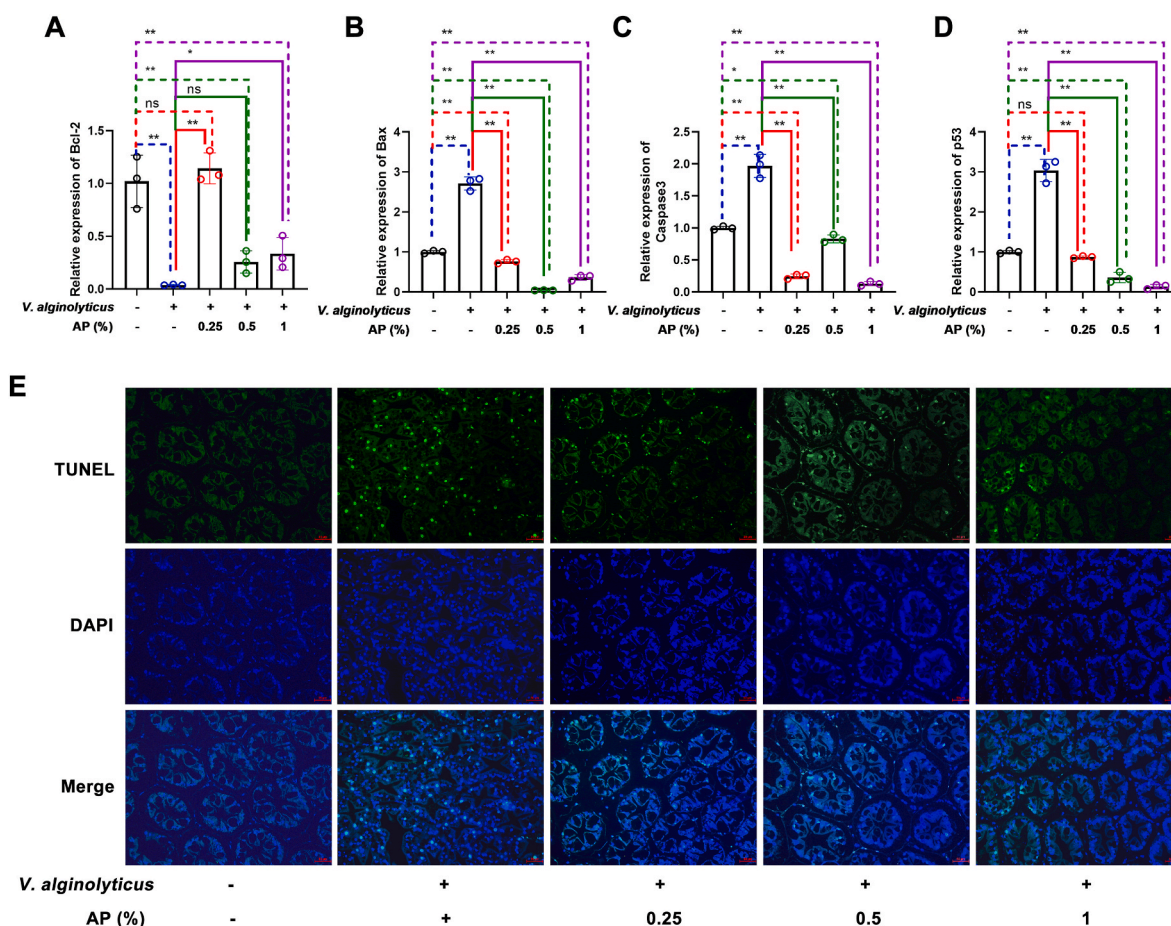


Fig. 10. Effects of AP on apoptosis-related genes Bax (A), Caspase3 (B), p53 (C) and Bcl-2 (C) in hepatopancreas of *L. vannamei* under the stress of *V. alginolyticus*. (D) Detection of apoptosis in hepatopancreas of *L. vannamei* by TUNEL stain after *V. alginolyticus* infection 6 h. Results were shown as mean $\pm$ SD. Statistical differences were evaluated by a one-way ANOVA, where ns means $p > 0.05$, * means $p < 0.05$, ** means $p < 0.01$.

influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

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